

MCQs on Shell Programming

1. What is the purpose of the if-else statement in shell programming?

- A) To create loops
- B) To execute a block of code based on a condition
- C) To define functions
- D) To perform arithmetic operations

Answer: B) To execute a block of code based on a condition

2. Which command is used to evaluate conditions in shell scripts?

- A) if
- B) test
- C) case
- D) while

Answer: B) test

3. What does the following syntax represent? `if [ condition ] then commands else commands fi`

- A) A loop
- B) A function
- C) A decision-making statement
- D) A regular expression

Answer: C) A decision-making statement

4. In a nested if-else statement, what is true about the structure?

- A) It cannot contain more than two conditions.
- B) It can have an if statement within another if block.
- C) It must always have an else clause.
- D) It is only used for arithmetic operations.

Answer: B) It can have an if statement within another if block.

5. What is the syntax for a case statement in shell programming?

- A) `if [ condition ]; then commands; fi`
- B) `case variable in pattern1) commands ;; pattern2) commands ;; *) default_commands ;; esac`
- C) `for variable in list; do commands; done`
- D) `while [ condition ]; do commands; done`

Answer: B) `case variable in pattern1) commands ;; pattern2) commands ;; *) default_commands ;; esac`

6. Which loop repeats commands as long as a condition is true?

- A) until loop
- B) for loop
- C) while loop
- D) case loop

Answer: C) while loop

7. What does the following command do? `until [ $count -gt 5 ] do echo "Count: $count" count=$((count + 1)) done`

- A) It counts from 1 to 5.
- B) It counts until the count is greater than 5.
- C) It counts down from 5 to 1.
- D) It does not execute any commands.

Answer: B) It counts until the count is greater than 5.

8. How do you define a for loop that iterates through a sequence of values?

- A) for variable in list; do commands; done
- B) while [condition]; do commands; done
- C) until [condition]; do commands; done
- D) case variable in pattern; do commands; done

Answer: A) for variable in list; do commands; done

9. Which of the following is a valid regular expression syntax to match any single character?

- A) *
- B) .
- C) []
- D) ^

Answer: B) .

10. What does the command `grep "^Start" file.txt` do?

- A) Matches lines ending with "Start".
- B) Matches lines starting with "Start".
- C) Matches any line containing "Start".
- D) Matches lines that do not contain "Start".

Answer: B) Matches lines starting with "Start".

11. How do you perform arithmetic operations in shell scripting?

- A) Using the `arithmetic` command

- B) Using the `expr` command
- C) Using `$((expression))`
- D) Using `calc` command

Answer: C) Using `$((expression))`

12. What will the following code output if the user inputs 5?

```
bash
VerifyOpen In EditorRunCopy code
1read -p "Enter a number: " num
2for i in {1..10}
3do
4  echo "$num x $i = $((num * i))"
5done
```

- A) A multiplication table for 5
- B) An error message
- C) A countdown from 5
- D) Nothing

Answer: A) A multiplication table for 5

13. What does the command `wc -w < $file` do?

- A) Counts the number of lines in a file
- B) Counts the number of characters in a file
- C) Counts the number of words in a file
- D) Displays the content of the file

Answer: C) Counts the number of words in a file